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Balancing the pretreatment effect of deep eutectic solvents on biomass and the activity of β-glucosidase originating from *Paenibacillus* sp. LLZ1

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ABSTRACT

The two major impediments to biomass resourcing are the catalytic efficiency of enzymes and the effectiveness of biomass pretreatment. In this study, six genes encoding β-Glucosidases (BGLs) derived from *Paenibacillus* sp. LLZ1 were cloned and expressed. The activity of BGL-F, which had the highest enzyme activity, was 24.1 U/mg, which was 39.8–83.7% higher than that of the other BGLs. Molecular docking and surface charge analyses showed that the binding energy of BGL-F to cellobiose and its electronegative ligand-binding pocket were greater than those of other BGLs. On the other hand, by calculating and comparing the Kamlet–Taft (KT) values of deep eutectic solvent (DES) composed of different hydrogen bond donors (HBD) and hydrogen bond acceptors (HDA), it was found that DES composed of betaine and glycerol had the highest α value of 0.97, meaning that it could provide the highest hydrogen bonding capacity, and thus it had the best pretreatment effect on biomass. Analysis by circular dichroism (CD) spectroscopy revealed a decrease in α-helix and an increase in β-strand in the secondary structure of the DES-treated BGL-F, implying an increase in enzyme activity and stability. This result was also confirmed experimentally by the fact that the K_m and V_{max} of the enzyme in the DES system increased by 13.2% and 57.9%, respectively, and the inactivation rate (k_d) decreased by 21.4%. The results of this study showed that DES consisting of glycerol-betaine achieved a better balance between the promotion of enzyme activity and biomass pretreatment.

1. Introduction

With rapid economic development, people's demand for resources is increasing [1]. Among all resources, lignocellulosic biomass is the most abundant and recyclable resource in nature [2]. However, since the hemicellulose and lignin in lignocellulose are wrapped around the cellulose by covalent and hydrogen bonds forming a tight structure [3], it is not sufficient to enhance the enzyme activity alone. Therefore, finding a way to break the hydrogen bonds and van der Waals forces that form between hemicellulose and lignin, thus making it easier for enzymes to access cellulose, has become an urgent problem to solve [4]. This problem can be solved by pre-treating the biomass by deconstructing the lignified cell walls and furthering the effectiveness of the biomass conversion process [5]. Currently, research is devoted to the study of various solvents for decomposing biomass, which are not only efficient in treating biomass, but also environmentally friendly [6,7].

Among the most commonly used solvents for biomass pretreatment are ionic liquids [8–10], however, their fundamental drawbacks are their high cost and the difficulties in their dissemination and industrial application due to the complexity of the synthesis process [11]. Recently, deep eutectic solvent (DES), a eutectic mixture of hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA), has emerged as an emerging method for biomass pretreatment. DES is highly advantageous in biomass pre-treatment, as it is not only cost-effective, easy to synthesize, and biodegradable, but also environmentally friendly and non-combustible [12–14]. During biomass pretreatment, DES effectively separates lignin, hemicellulose, and cellulose and leaves cellulose intact without severe degradation. The characteristics of the wall polymer are significantly altered during the pretreatment process, thus greatly increasing the enzymatic accessibility of cellulose [15,16]. The hydrogen bonding acidity (Kamlet-Taft α parameter) of DES is positively correlated with the effect of pretreatment [7], which elaborates the relationship between the structure and performance of DES so that the DES with the best effect of cellulose pretreatment can be rationally designed. Meanwhile, it has been reported that laccase activity can even be increased by 200 % in specific DES systems [17]. So, it is a valuable research direction to search for a DES that can enhance the enzyme activity as well as be effective for biomass pretreatment. Presently, DES

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Table 1
Primers used in this study.

Primers	Sequences (5′→3′)	Enzyme digestion sites	Target gene size (bp)
BGL-A F	CGGAATTCCGAGGTAATCTAAACGAGC	EcoRI	2127
BGL-A R	CCCAAGCTTCTTGCTGTTAGGTCCGACG	HindIII	
BGL-B F	CGCGGATCCATGACGAAGTACGAATATCCG	BamHI	2811
BGL-B R	CCGGAATTCTTAAACCAGTTTCAACCAGCTG	EcoRI	
BGL-C F	CGCGGATCCATGACGAAGCCTACGCTATC	BamHI	2643
BGL-C R	CCGGAATTCCTACACCACTGTAATCTCCGC	EcoRI	
BGL-E F	CCGGAATTCATGGCAATTGTACAGTTTC	EcoRI	1347
BGL-E R	CCCAAGCTTTTATATTGTGAACCAGTTC	HindIII	
BGL-F F	CGCGGATCCATGGATCTATTCGGAGAGG	BamHI	1374
BGL-F R	CCCAAGCTTGTTATTCAATTGTGTTAGCCG	HindIII	
BGL-G F	CGCGGATCCTTGAGTACTGACAAAACAATTG	BamHI	2775
BGL-G R	CCCAAGCTTTTAGATTTGCTGGAACTCAATG	HindIII	

commonly used for biomass pre-treatment often uses choline chloride (ChCl), betaine, or benzyltrimethyl-ammonium chloride (BTMAC) as HBA [18–20]. It has been demonstrated that HBD with urea-based DES improves the stability of the enzyme in the reaction system, and glycerol-based DES also promotes enzyme activity in the reaction system [21–24]. Nevertheless, studies of conformational changes of enzymes in DES systems are lacking, and investigating the reasons for the increased stability of enzymes in DES could provide further guidance for DES selection.

In this study, *Paenibacillus* sp. LLZ1 (*P. sp.* LLZ1) previously screened by our group was selected for culture [25,26], and the BGL gene from it was cloned and expressed in *E. coli* BL21. Differences between different genes encoding the same class of enzymes, as well as differences in the enzymatic properties of these enzymes, were studied and compared to identify and screen the enzymes with the highest enzymatic activity and stability. Specifically, the stability of the enzymes in different DES was determined. The time course of substrate hydrolysis, enzyme kinetics, and kinetics of inactivation under different DES-treated biomass were also determined. Finally, the secondary structure of the DES-treated enzyme was characterized by circular dichroism to determine the effect of DES solvent on the conformation of the enzyme. Herein, the effects of different DES on the pretreatment of biomass and enzyme activity were investigated to find the equilibrium between the two, and the effects of DES on the secondary structure of the enzyme were also investigated to elucidate the mechanism of DES-enhanced enzyme activity.

2. Materials and methods

2.1. Materials

p-nitrophenyl-β-D-glucopyranoside (*p*NPG) and *p*-Nitrophenol(*p*NP) were purchased from Aladdin (Shanghai, China). Plasmid extraction kits and one-step PAGE gel fast preparation kits were purchased from Vazyme Biotech (Nanjing, China). BeyoGold His-tag purification resin with 50 % concentration was purchased from Beyotime (Shanghai, China). Choline chloride, BTMAC, betaine, urea, glycerol, and other reagents were purchased from Aladdin (Shanghai, China).

2.2. Cloning of the BGL gene

The whole-genome sequencing of *Paenibacillus* sp. LLZ1 was performed by Beijing Tsingke Biotech Co., Ltd., resulting in six genes encoding BGL. The six gene sequences were used as templates to design primers according to the primer design principle, and enzyme cleavage sites were added to the 5′ and 3′ ends of the primers respectively. The results of the primer design are listed in Table 1. The six genes were then amplified by polymerase chain reaction (PCR). The target genes obtained by PCR amplification were separated by 1 % agar gel electrophoresis in TAE buffer solution and purified using gel recovery for target gene isolation. The pET-22b (+) plasmid and the target gene were double cleaved with three endonucleases, EcoRI, HindIII, and BamHI. T4 ligase was used to connect the end of the target gene and the plasmid, which were cut with the same endonuclease. The final transformation into *E. coli* BL21 receptor cells were carried out by thermal excitation and cultured on Luria-Bertani (LB) agar plates containing 50 μg/ml of ampicillin.

2.3. Expression and purification of six BGLs

The already constructed successful strain preserved in glycerol was inoculated in Luria-Bertani (LB) medium containing ampicillin. Activation was carried out by incubation at 37 °C, 200 rpm for 12–16 h. Then 1 % of the culture was inoculated in a new ampicillin-containing LB medium. At an OD_{600} of approximately 0.6, 0.4 mM Isopropyl β-D-Thiogalactoside (IPTG) was added to the medium, and protein expression was induced by incubation at 20 °C, 200 rpm for 20 h. The supernatant was taken and centrifuged at 4 °C, 8000 rpm for 20 min, and washed several times with PBS. Then the bacterial cell spheres were sonicated at 4 ℃, and after the solution became clear and transparent, the supernatant was harvested by centrifugation at 10,000 rpm for 20 min at 4 ℃. The Ni^{2+}-NTA agarose gel column was used for purification. The bound recombinant protein was eluted with a gradient of imidazole solution, and then concentrated by centrifugation at 4 ℃ and 6000 rpm using an ultrafiltration tube and stored at 4 ℃. The concentration of purified protein was determined by the Bradford method using bovine serum albumin (BSA) as a standard. The molecular weight of the enzyme and the purification process were evaluated using SDS-PAGE electrophoresis.

2.4. Enzyme assays

The enzymatic activity of BGLs was measured using *p*-nitrophenyl-b-D-glucopyranoside (*p*NPG) as substrate. The previous assay protocol was continued [27]. 3 g/L (6.65 mM) *p*NPG and 1 g/L BGL were added to the reaction system and buffered with PBS (pH 7.0). The reaction was terminated by the addition of 1 M Na_2CO_3 after 1 h in a water bath shaker at 50 °C. The release of the product yield *p*-nitrophenyl (*p*NP) was determined using a colorimetric assay, and the amount of reaction product was detected at 405 nm absorbance using a 96-well plate (Sunrise; Tecan Trading AG, Männedorf, Switzerland). All data are averages of three experiments. One unit (U) of BGL activity was defined as the amount of enzyme that liberates 1 μmol *p*NP per minute under optimum assay conditions. The concentration of *p*NP was calculated from the standard curve.

The optimum temperature and optimum pH of the enzyme were determined using *p*NPG as substrate. The optimum temperature and optimum pH of the different BGLs were measured by incubating the enzyme for 1 h at 200 rpm in a buffer with a series of pH gradients from

Table 2
Deep eutectic solvents (DES) preparation.

DES	molar ratios	Temp.	Time	Ref.
ChCl-urea	1:2	80℃	30 min	[16]
ChCl-glycerol	1:2	80℃	30 min	[17]
ChCl- glycerol-urea	1:1:1	80℃	30 min	[36]
BTMAC-glycerol	1:2	65℃	1 h	[18]
BTMAC-glycerol-urea	1:1:1	65℃	1 h	[18]
Betaine-glycerol	1:2	80℃	30 min	[37]
Betaine- glycerol-urea	1:1:1	80℃	30 min	[38]

5.0 to 10.0 and a series of temperature gradients from 20 to 70 °C.

2.5. Kamlet–Taft parameters measurement of DESs

The Kamlet–Taft (KT) parameters of DESs were measured using the solvatochromic dyes of 4-nitroaniline (4NA), *N, N*-diethyl-4-nitroaniline (DENA), and Reichardt's dye (RD). First, 2 mg of each dye was dissolved in 4 ml of ethanol. Then the ethanol was evaporated at 40 °C for 12 h. DESs were added and stirred to redissolve the dyes. The UV–visible spectra of the dyed DESs were measured in the range of 350–700 nm at 25 °C using a UV–visible spectrophotometer in a quartz cell with a light-path length of 1.0 mm. The correlation data were used to calculate bipolarity-polarizability (π^*), hydrogen bond acidity (α), and hydrogen bond basicity (β),[28]

$$\pi^* = \frac{\nu_{max,DENA} - 27.52}{-3.182} \quad (1)$$

$$\alpha = \frac{E_T(30) - 14.6(\pi^* - 0.23) - 30.31}{16.5} \quad (2)$$

$$\beta = \frac{1.305\nu_{max,DENA} - \nu_{max,4NA} + 2.64}{2.8} \quad (3)$$

in which ν_{max} is related to the wavelength of maximum absorbance of its corresponding dyes, $\nu_{max} = 10^4/\lambda_{max}$. $E_T(30)$(kcal mol^{-1}) indicates solvent polarity, $E_T(30) = 28591/\lambda_{max,RD}$.

2.6. DES preparation and bagasse pretreatment

ChCl, btmac, and betaine as hydrogen bond acceptors and glycerol, and urea as hydrogen bond donors were heated and stirred at 80 °C for one hour. Once a clear homogeneous liquid was obtained it was cooled in a desiccator to prevent moisture absorption. The specific preparation ratios are listed in Table 2.

Using bagasse biomass as an enzymatic substrate, it needs to be pretreated using DES before in situ saccharification. Firstly, the biomass was washed three times each using anhydrous ethanol and Deionized water (DIW) respectively, dried at 55 °C, and set to constant weight. Next, the biomass for use in the reaction was obtained by pulverizing the process using a high-speed universal pulverize and passing through a 40-mesh sieve. The biomass powder was thoroughly mixed with DES and kept in a boiling water bath for 10 min until the color and fluidity of the mixed system no longer changed, and the buffer was added to cool it down to complete the pretreatment. In addition, the amounts of DES and buffer were calculated and added according to the concentration of each substance and system capacity in each experiment.

2.7. Enzymatic hydrolysis

Microcrystalline cellulose was first used as a substrate to determine the product yields at different ratios of cellulose and BGL under optimum reaction conditions. After measuring the optimum ratios, the reaction was carried out under optimum reaction conditions using sugarcane bagasse pretreated and unpretreated through various DESs as substrates. The yields were determined by the (3,5-dinitrosalicylic acid) DNS colorimetric method and the pretreatment effect of each DES solution was further evaluated.

Table 3
GH families to which BGLs belong.

Enzyme	GH	Source strains of BGL with highest similarity	GeneBank	Ref.
BGL-A	1	*Paenibacillus castaneae*	NIK75795.1	[30]
BGL-B	3	*Paenibacillus terricola*	WP_191205513.1	[31]
BGL-C	3	*Paenibacillus cellulosilyticus*	WP_317972457.1	[31]
BGL-E	1	*Paenibacillus terricola*	WP_191206889.1	[31]
BGL-F	1	*Paenibacillus cellulosilyticus*	WP_317964914.1	[32]
BGL-G	3	*Paenibacillus terricola*	WP_191206634.1	[31]

2.8. Determination of kinetic parameters

The initial reaction rate of the cellulase-BGL complex enzyme preparation in the presence of different DES was determined using MCC as a substrate to determine the kinetic parameters. V_{max} and K_m values were calculated using Michaelis-Menten nonlinear regression.

In addition, the time-loss-of-activity kinetics of the cellulase-BGL complex enzyme preparations in different DES were determined. The residual activity (%) at certain time intervals was calculated by determining the amount of *p*NP produced by hydrolysis of the cellulase-BGL complex enzyme preparation at optimal pH and temperature. The initial activity at the start of the reaction was set at 100 %. The mechanics of inactivation were determined by Eq. (4)[29].

$$a = (a_0 - a_p)e^{-k_d \bullet t} + a_p \quad (4)$$

where (a), (a_0), (a_p), and (k_d) denote residual activity, initial enzyme activity, final enzyme plateau activity, and equilibrium dissociation constant, respectively. In particular, (a_p) is associated with concentration-dependent equilibrium, and both it and (k_d) are key data for measuring enzyme stability. High (a_p) and low (k_d) indicate that the enzyme is less likely to be inactivated in the reaction system.

2.9. Characterization

The proteins used for characterization were kept in a 25 % concentration of different DES for 12 h. The DES was then removed from the solution by dialysis and the enzyme was diluted to 0.5 g/L using ddH_20 as a dispersing medium.

Circular dichroism spectra (CD spectra) were measured using a Circular Dichroism Spectrometer (JASCO J-1500, JPN.) at 25 °C in the wavelength range of 180–300 nm, and the measured protein secondary structure data were analyzed using CDpro software.

3. Results and discussion

3.1. Heterologous expression Paenibacillus sp. LLZ1 BGLs

The total genome of *Paenibacillus* sp. LLZ1 BGLs were used as a template to obtain PCR products using *Taq* DNA polymerase, which was transfected into Escherichia coli PET-22b receptor cells after digestion and enzyme linkage. Measurement results showed that six BGL genes with lengths ranging from 1347 bp to 2811 bp were inserted in each of the six groups of receptor cells. A search of the database by the BLAST programmer revealed that all six amino acid sequences shared 97–99 % homology with the BGL in *Paenibacillus terricola* (Genebank: WP_191204150.1) or *Paenibacillus cellulosilyticus* (Genebank: GMK37628.1). The GH families to which BGLs belong are listed in Table 3.

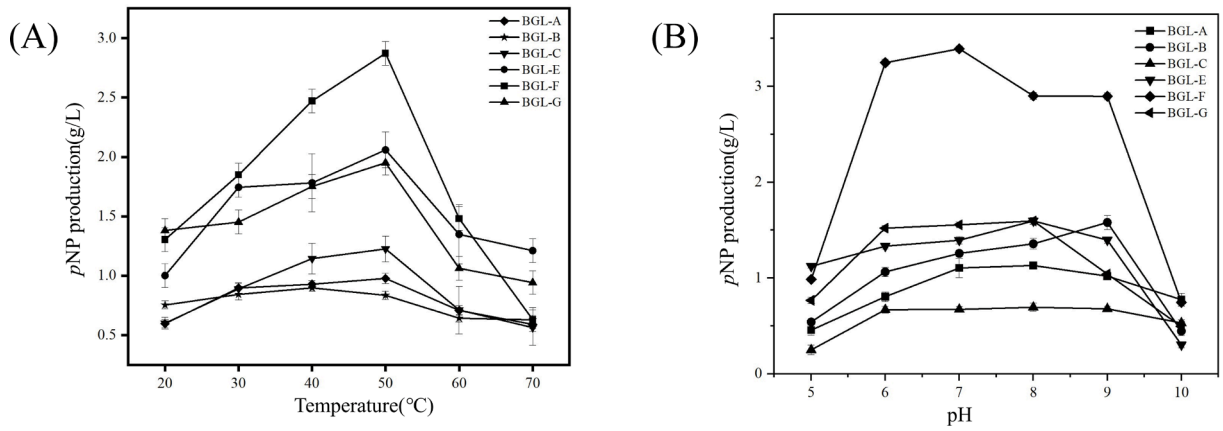


Fig. 1. (A)The optimum temperature of different BGLs originating from *Paenibacillus* sp. LLZ1. (B)The optimum pH of different BGLs originating from *Paenibacillus* sp. LLZ1.

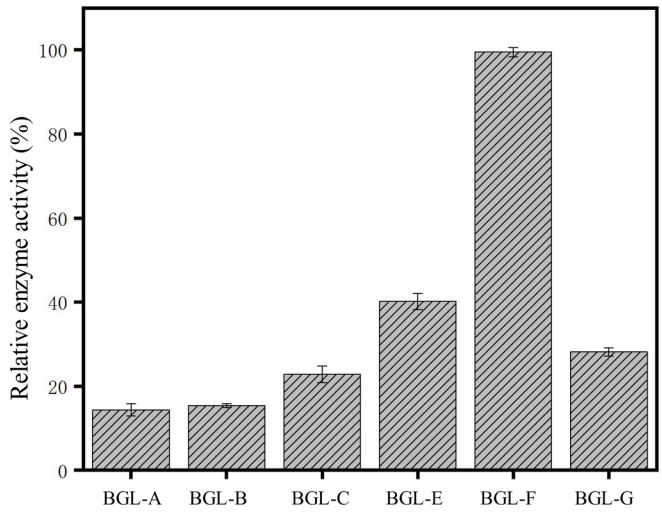


Fig. 2. Relative catalytic efficiency of different BGLs originating from *Paenibacillus* sp. LLZ1.

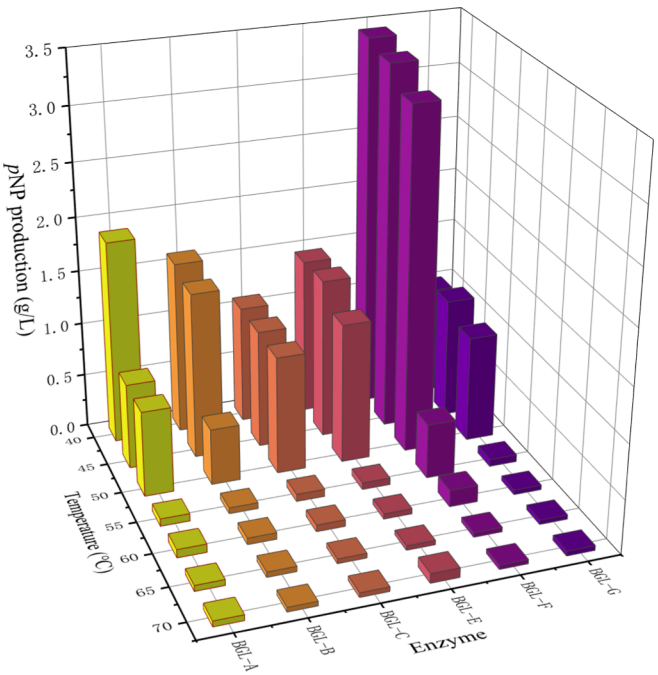


Fig. 3. Thermal stability of different BGLs originating from *Paenibacillus* sp. LLZ1.

3.2. Enzymatic properties of BGLs

The temperature and pH dependence of the six BGLs were determined (Fig. 1), and the best performer was BGL-F. Moreover, the optimum temperature for all six BGLs was 50 °C, and the optimum pH was neutral or slightly alkaline. The specific enzyme activity of BGLs was determined at 50 °C and pH = 7, and the specific enzyme activity of BGL-F was 24.1 U/mg, which was 39.8 ~ 83.7 % higher than that of the other enzymes (Fig. 2). In addition, the heat resistance of the enzyme was also explored to some extent, and it was learned that the heat resistance of BGL-F also has a great advantage compared to other enzymes (Fig. 3). To investigate the reason for the superior nature of BGL-F over other enzymes, six BGLs were molecularly docked using fiber disaccharide, the real substrate of BGL, as the substrate (Fig. 4, Fig. 5). In terms of the binding energy of the BGLs molecules to the fibrous disaccharide, the binding of all the enzymes to the substrate is an exergonic reaction, and the absolute value of the binding energy of BGL-F is the largest (Table 4) so that it achieves a more stable state than the rest of the enzymes after binding to the substrate [33]. In terms of the number of hydrogen bonds formed, BGL-F formed 2 hydrogen bonds with fiber disaccharide, which is also one of the important reasons for its stable binding state. After molecular docking, it was found that the binding pocket of BGL-F consisted of hydrophilic amino acids such as Glu-327, and His-321, whereas the other enzymes had a smaller number of hydrophilic amino acids. It has been shown that hydrophilic amino acids are an important part of the binding process between BGL and substrate, especially Glu, which both breaks the glycosidic bond and participates in the formation of the end product glucose [34,35].

By analyzing the charge distribution model of the enzyme (Fig. 6), it was found that the negative pocket of BGL-F was larger than that of the rest of the enzymes, while the ligand small molecules were bound by the negative pocket, the larger binding pocket was more conducive to substrate entry [36], which was one of the important reasons why the activity of BGL-F was superior to that of the other enzymes. Selection of BGL-F for subsequent reactions.

3.3. Effectiveness of DES solution in pre-treatment of biomass

It is well known that an important impediment to cellulose hydrolysis in biomass is the haphazard arrangement of hemicellulose and lignin. The breaking of various aryl ether and ester bonds in lignin can be promoted in DES to facilitate the solubilization of lignin [20]. The

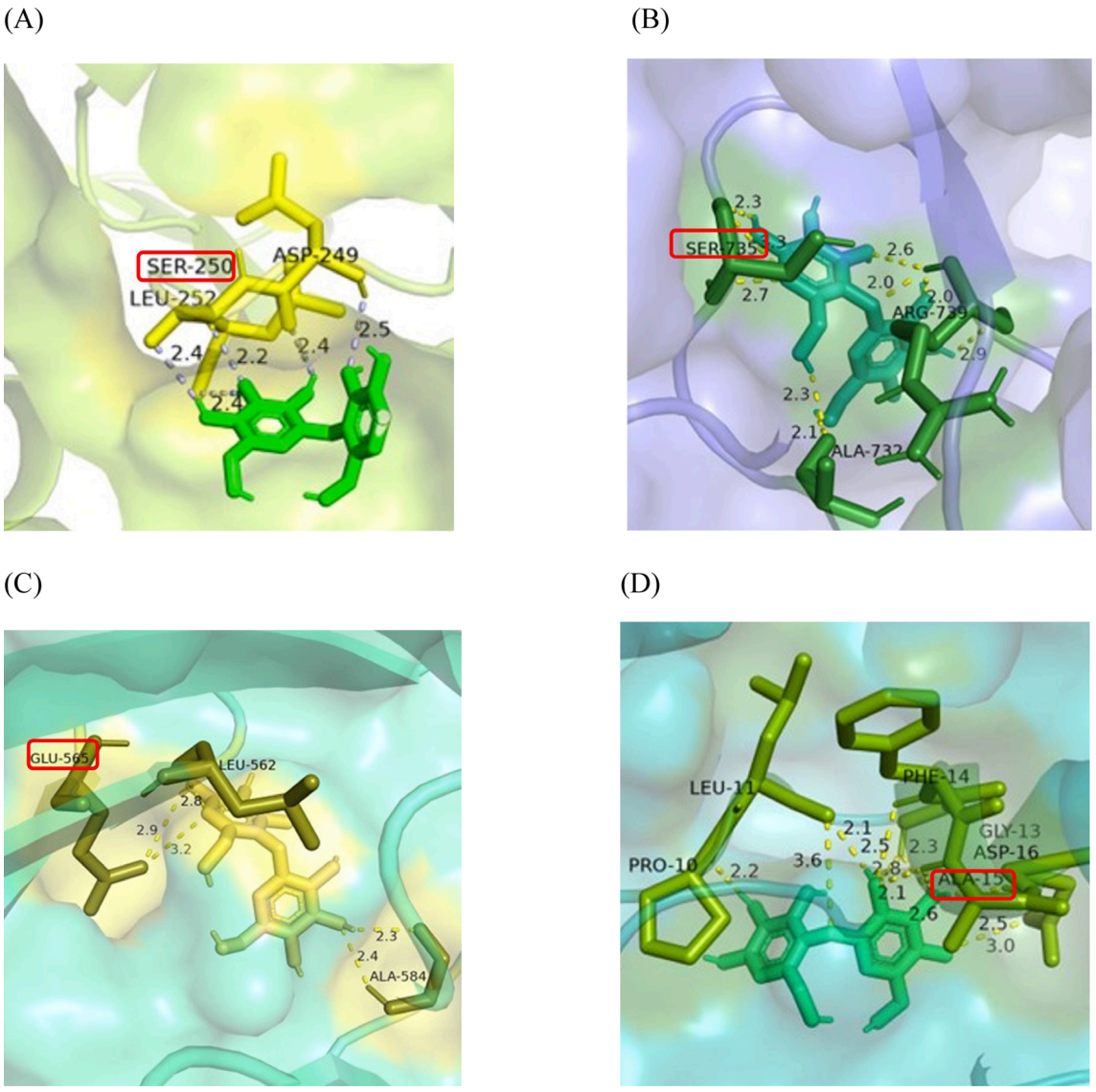


Fig. 4. Molecular docking results for different BGLs originating from *Paenibacillus* sp. LLZ1 using cellobiose as substrate.

reason is that the hydrogen bond between OH/Cl in DES is stronger than the total OH/O bond of lignin, and the presence of a large amount of Cl- enables DES to cleave the lignin-carbohydrate complex (LCC) in biomass as well as some ether bonds in lignin. Hemicellulose can be effectively removed by crushing of LCC to improve enzyme accessibility and hydrolysis rate [37].

Besides, the changes in the relative enzyme activities of the compounded enzyme preparations for the catalytic hydrolysis of MCC at cellulase: BGL ratios of 2:1, 1:1, 1:2, 1:5, and 1:10 were investigated, respectively. As the content of BGL gradually increased, the relative enzyme activity of the formulated enzyme showed an increasing and then decreasing trend, reaching a peak at 1:2. This is because the additional addition of BGL would further hydrolyze off the accumulated cellobiose in the enzymatic pathway to a certain extent, attenuating the negative feedback inhibition. However, as the proportion of cellulase in the synergistic system becomes lower and lower, the upstream ability to gradually decompose cellulose chains into cellobiose becomes weaker and weaker, and then even if there is more BGL in the reaction, it will not help. Therefore, the optimal ratio of C: BGL in the subsequent synergistic enzymatic system was determined to be 1:2.

As shown above, we pretreated bagasse with seven different DES. Using those differently pretreated bagasse as substrate, the reaction was carried out by adding the optimum proportion of cellulose-BGL complex enzyme preparation under optimum conditions. Optimal reaction in the presence of DES prepared with glycerol as HBD and betaine as HBA in the reaction system (Fig. 7). This may contain two reasons: (1) the good hydrolysis of biomass by DES prepared with glycerol as HDA and betaine as HBD. (2) The DES has a preserving or even enhancing effect on the activity and stability of the enzyme.

The polarity of DESs is quantified by the Kamlet-Taft parameter, where α, β, and π^* represent the offered hydrogen bonding capacity, the accepted hydrogen bonding capacity, and the polarity/polarization ratio, respectively. The α and β values of DES are closely related to the biomass pretreatment effect, with the α value being proportional to the H^+ released during DES preparation and the β value being proportional to the number of hydrogen bonds formed [38]. Table 5 lists the α and β values of various DES with glycerol as the HBD, and it was found that the α value with betaine as the HBA was the highest. This is because betaine contains not only nitrogen ions but also carboxylic acid groups, which have strong electron-accepting and electron-giving abilities [38]. Therefore, when glycerol was used as the HBD, the DES prepared with betaine as the HBA had the best pretreatment effect.

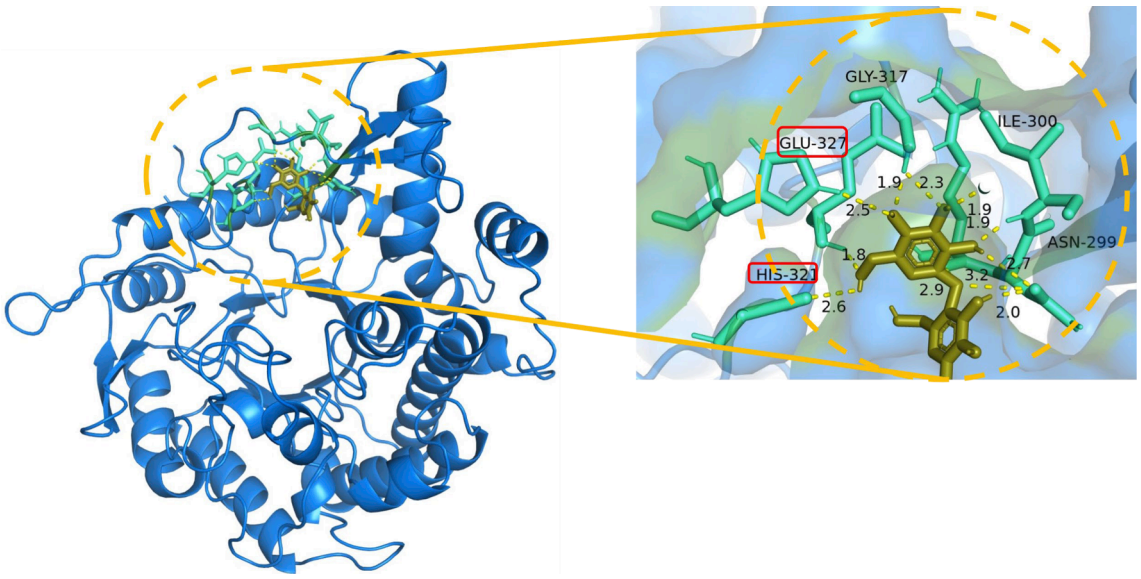


Fig. 5. Molecular modelling and docking results for enzyme F.

Table 4
Molecular docking results of BGLs with cellobiose.

Enzyme	BGL-A	BGL-B	BGL-C	BGL-E	BGL-F	BGL-G
Binding energy (kcal/mol)	−4.19	−4.54	−3.77	−5.63	−7.83	−5.96
Hydrogen bond	1	1	0	1	2	3

3.4. Enzymatic kinetic parameters in the DES system

First, we used biomass pretreated with different DESs as substrates for enzymatic hydrolysis using a cellulase-BGL combination enzyme system. The results of the experiments are shown in Fig. 7, the highest yield of glucose was obtained by enzymatic hydrolysis after DES pretreatment prepared from glycerol-betaine. In contrast, reducing sugars were detected in the unpretreated biomass, indicating that only a small portion of the cellulose in the biomass could be exposed to the surface of the biomass under unpretreated conditions. Similarly, in the experiment where only the biomass was pretreated without the addition of enzyme, a small portion of reducing sugar was still produced, indicating that the DES solvent has some ability to dissolve cellulose [28]. The catalytic reaction kinetics (Michaelis-Menten equation) of the cellulase-BGL complex enzyme preparation (Fig. 8A) showed that V_{max} and K_m were higher with increasing substrate concentration and with the presence of DES in the reaction system (Table 6). Compared to the V_{max} (5.06 g/L/h) and K_m (0.19 g/L) of the enzyme in PBS buffer, the enzyme had higher V_{max} (5.73 g/L/h, 5.19 g/L/h) and higher K_m (0.30 g/L, 0.21 g/L) with the addition of Betaine-glycerol, Btmac-glycerol. It indicated that the addition of DES was more conducive to the full contact between the enzyme and the substrate, which was beneficial to reducing the non-competitive inhibition of the enzyme by the substrate [29]. When

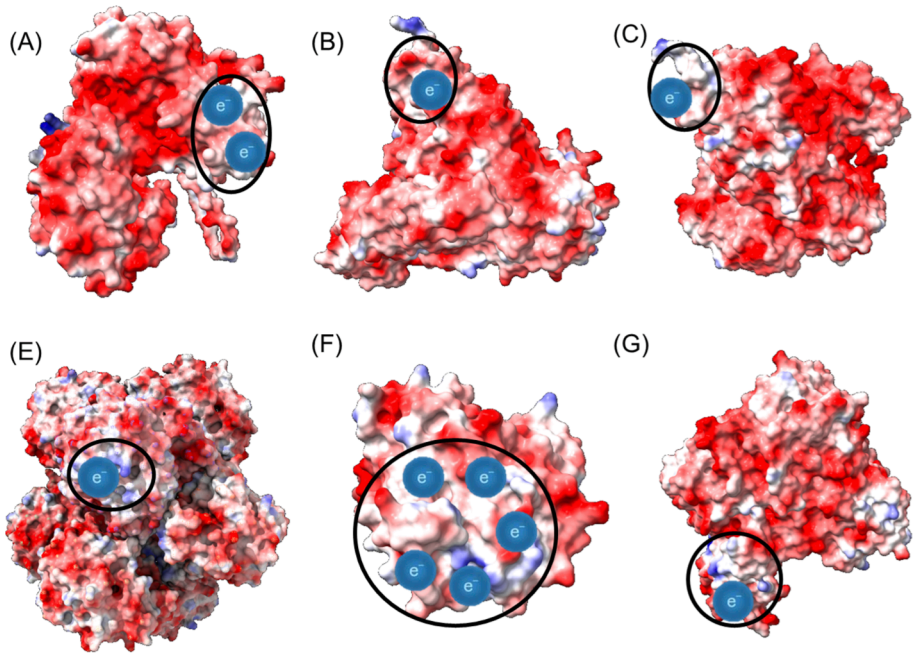


Fig. 6. Surface charge distribution of different BGLs.

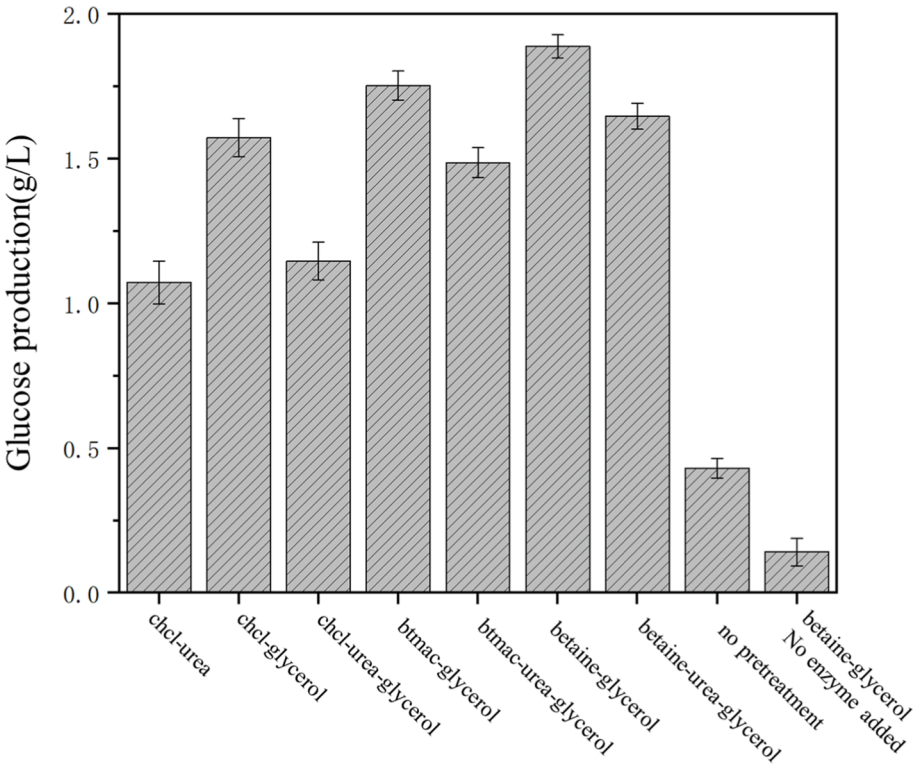


Fig. 7. Enzymatic degradation of biomass after pretreatment with different DESs.

Table 5
Kamlet–Taft parameters for different DESs.

NO.	HBD	HBA	Kamlet–Taft parameters		
			π*	α	β
1	Glycerol	Choline chloride	1.08	0.86	0.65
2		Betaine	1.06	0.97	0.62
3		Btmac	1.117	0.953	0.574

betaine binds to polyols to form DES, it may not react synergistically with the polyols, which effectively prevents protein inactivation and aggregation [39]. Another possible reason is that polyols have a stronger force to form hydrogen bonds than urea, which makes it easier to release the substrate from the hydrogen bonding network, thus facilitating the binding of the substrate to the active center of the enzyme [17].

Cellulase-BGL combination enzyme systems have a crucial role in the enzymatic saccharification of biomass, and most of the previous studies have considered only one of the enzymes, and there is a lack of research on combination enzyme systems of both enzymes. Based on previous studies [27], the time-loss-of-activity kinetics of hydrolysis of MCC by the cellulase-BGL combined enzyme system was investigated. As shown in Fig. 8(B), the combined enzyme system was sharply deactivated during the enzymatic reaction initially and reached equilibrium in about one hour, which is in agreement with previous studies [40]. The

Table 6
Enzyme kinetic parameters.

Sample	V_{max}(g·L^{-1}·h^{-1})	K_m(g·L^{-1})	R^2
PBS	5.06	0.19	0.9266
Btmac- glycerol	5.19	0.21	0.9498
Betaine-glycerol	5.73	0.30	0.9595

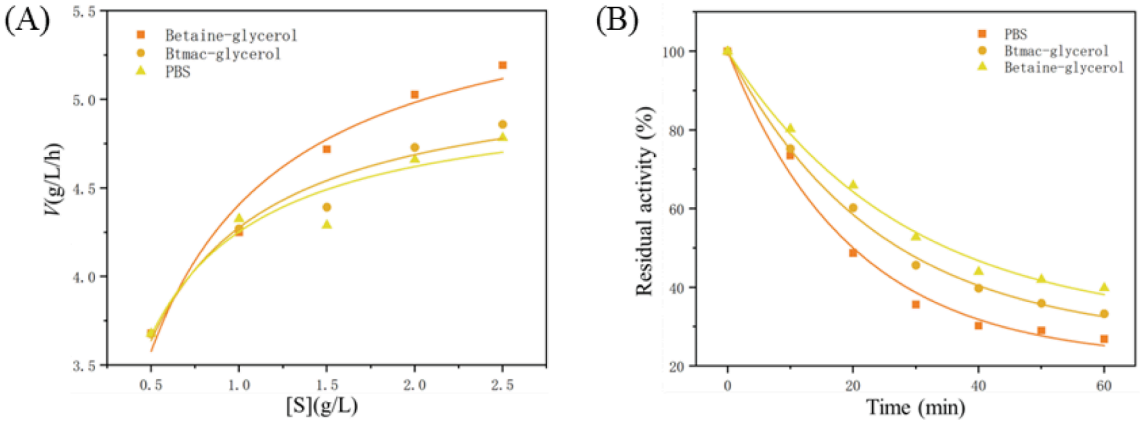


Fig. 8. (A)Enzymatic kinetic parameters of cellulase-BGL complex enzyme preparations in DES-involved enzymatic reactions (B)inactivation kinetics parameters of cellulase-BGL complex enzyme preparations in DES-involved enzymatic reactions.

Table 7
Parameterization of inactivity mechanics.

Sample	a_p (%)	k_d (min^{-1})	R^2
PBS	21.41	0.051	0.9914
Btmac- glycerol	26.50	0.042	0.9878
Betaine-glycerol	29.99	0.036	0.9947

combined enzyme system entirely in PBS buffer lost about 65 % of its activity in the first half hour of the reaction and only about 9 % in the second half hour. In contrast, after the addition of 25 % DES to the buffer, the combined enzyme system deactivated more slowly and had a higher final residual activity. The calculated deactivation kinetics parameters further illustrate the deactivation process (Table 7). Stable and less susceptible to inactivation enzymes exhibited higher final plateau activity (a_p) and lower rates of inactivation (k_d). The k_d of the combined enzyme system in DES containing was 0.042 min^{-1}, and 0.036 min^{-1} respectively, which were lower than that of the combined enzyme system in pure PBS buffer (0.051 min^{-1}). This suggests that they inactivate more slowly, which is related to the biocompatibility of betaine [39].

3.5. Circular dichroism spectral characterization

CD spectroscopy is an excellent technique commonly used to analyze changes in the secondary structure of proteins. The protein structure was examined at 180–300 nm and the results are shown in Table 8 and Fig. 9. In the buffer of DES prepared from betaine and glycerol, the amount of α- helix in the secondary structure of proteins decreased by 1.1 %, while the amount of β-folding increased by 1.6 %. It has been proved that the decrease in the content of α-helix and the increase in the content of β-folding will affect the active site of the enzyme, which will increase the enzyme activity [41].

In addition, the peak shape at 200–240 nm did not change after adding DES, but the signal was enhanced. It has been shown that the enhancement of signal is known to demonstrate the enhancement of protein stability. In particular, glycerol with three hydroxyl groups may bind to the protein by forming more hydrogen bonds, thus making the protein structure more rigid and stable [42].

4. Conclusion

The *P. sp.* LLZ1 BGLs previously screened by our group were subjected to whole genome sequencing, and six different BGL genes originating from Bacillus-like bacteria were cloned. E. coli expressing six different BGLs were constructed on this basis. The enzyme activities of the six proteins were determined, and one of them was screened to have a relative enzyme activity of 24.1 U/mg, which was 39.8–84.7 % higher than that of the other enzymes. And its thermal stability also has a big advantage. Molecular docking of the enzyme revealed that BGL-F was the most stable in binding to the substrate. Moreover, its binding pocket consists of hydrophilic amino acids such as Glu-327 and His-321, which are more conducive to substrate binding and product generation. Charge analysis of the enzyme revealed that BGL-F has more negatively charged groups than the rest of the enzymes, and the ligand small molecules are generally bound by negatively charged pockets, which is an important reason why the activity of BGL-F is superior to that of the rest of the enzymes. Subsequently, the effect of different fractions of DES on the pretreatment of biomass and the combined enzyme system of BGL-F and cellulase was investigated using sugarcane bagasse as substrate. By analyzing the correlation data (α, β, and π*) of DES, it was found that DES prepared when betaine was HDA and glycerol was HBD had a good pretreatment effect on biomass, which was in agreement with the experimental results. The enzymatic kinetics and inactivation mechanics parameters of the enzymes were determined and it was found that the DES solution consisting of betaine-glycerol is very biocompatible and promotes the activity of the enzymes. Finally, the DES-treated enzyme was characterized by circular dichroism spectroscopy, and it was found that the α-helix of the DES-treated enzyme was reduced and β-folding was increased. This also proved that the stability of the DES-treated enzyme was further increased.

CRediT authorship contribution statement

Fanjin Meng: Writing – original draft, Investigation. **Xin Ju:** Writing – original draft, Investigation. **Yuyang Zhou:** Supervision, Methodology. **Lishi Yan:** Writing – review & editing. **Zhi Chen:** Supervision, Methodology. **Yuqiu Wu:** Writing – review & editing. **Liangzhi Li:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

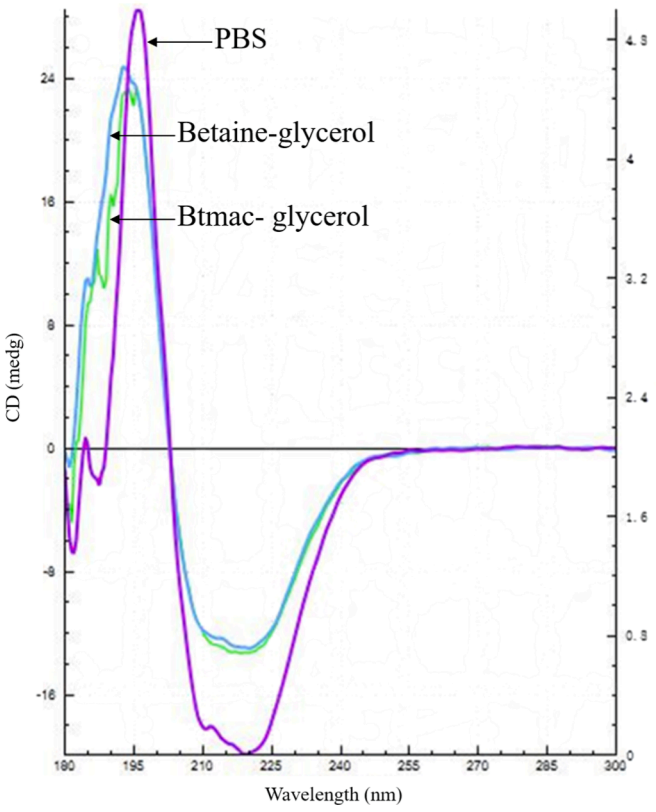


Fig. 9. Circular dichroism of BGL before and after DES treatment.

Table 8
Secondary structure content of BGL.

Buffer solution	Regular α-Helix	Distorted α-Helix	Regular β-Strand	Distorted β-Strand	Turns	Unordered
PBS	0.010	0.200	0.276	0.098	0.214	0.183
Btmac- glycerol	0.035	0.157	0.283	0.142	0.125	0.258
Betaine-glycerol	0.040	0.159	0.282	0.108	0.159	0.252

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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